Curriculum Vitae

DI ZHU

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University of Minnesota, Twin Cities — Cell: +1.612.206.6873
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Education

Ph.D., Peking University. *Cartology and Geographic Information Science* 2020
Title: Modelling and Analyzing Geospatial Distributions with Artificial Neural Networks.
Advisor: Dr. Yu Liu

B.S., Peking University. *Geographic Information Systems* 2014

B.Ec., Peking University. *Economics* 2014

Positions/Employment

University of Minnesota, Twin Cities

Assistant Professor, Department of Geography, Environment and Society 2021 - Present
[Start date delayed from Fall 2020 to Fall 2021 due to COVID-19 pandemic]

Affiliated Faculty, Minnesota Population Center 2022 - Present

Affiliated Faculty, CSE Data Science Initiative 2022 - Present

Scholar, Center for Transportation Studies 2025 - Present

Core Member, Data Science and AI Hub (DSAI) 2025 - Present

University College London

Visiting Lecturer, Civil Environmental & Geomatic Engineering 2018 - 2019

Peking University

Graduate Research Assistant, School of Earth and Space Sciences 2016 - 2020

Graduate Research Assistant, Geosoft Lab 2017 - 2018

Teaching Assistant, School of Earth and Space Sciences 2015 - 2018

Undergraduate Research Assistant, Geosoft Lab 2013

RS, GIS & GPS Practice Intern, School of Earth and Space Sciences 2012

Beijing GeekArt Technology Co. Ltd.

Chief Product Officer 2018- 2020

Beijing LongRuan Technology Co. Ltd

Software Engineer Intern 2015

Current Membership in Professional Organizations

International Association of Chinese Professionals in Geographic Information Sciences (CPGIS)	2017 - Present
Association of American Geographers (AAG)	2019 - Present
ACM Special Interest Group on Spatial Information (ACM SIGSPATIAL)	2022 - Present
University Consortium for Geographic Information Science (UCGIS)	2024 - Present
Association of Collegiate Schools of Planning (ACSP)	2025 - Present

HONORS AND RECOGNITION

(*Underline* - *student advisee awards*)

University of Minnesota

McKnight Land-Grant Professorship	2026
Mentor of Dean's First-Year Research & Creative Scholars (DFRACS), *College of Liberal Arts*	2025

External Sources

Top 20 Global Young Scientist (Finalist), *World Geospatial Developers Conference*	2022
Distinction of Doctoral Thesis, *Peking University*	2020
Excellent Graduates, *Peking University*	2020
China National Scholarship, *Ministry of Education, P. R. China* (CNY30,000)	2019
Early Career Scholarship, *GIS Research United Kingdom*	2019
AAG Travel Award, *Applied Geography Speciality Group* ($250)	2019
Rising Star Award, *College GIS Forum (CGF), China*	2018
Outstanding Student Award, *Peking University*	2018
Presidential Fellowship, *Peking University* (CNY60,000)	2018
Tang Lixin Scholarship, *Peking University* (CNY30,000)	2017
Individual Scholarship for Outstanding Scientific Research, *Peking University*	2015
Longruan Technology Award, Beijing LongRuan Technology (CNY5,000)	2014
Founder Scholarship, *Peking University* (CNY5,000)	2012
54 Scholarship, *Peking University* (CNY1,500)	2011

Advisee Honors and Recognition

Undergraduate Research Opportunities Program (UROP) Award, *Office of Undergraduate Research* (Undergraduate Student Awardee: Mathias Kreuziger)	2026
Waldo Tobler GISS-SG Student Competition, *Geographic Information Science & Systems Specialty Group, Association of American Geographers* (Finalist; Graduate Student Awardee: Meicheng Xiong)	2026
Waldo Tobler GISS-SG Student Competition, *Geographic Information Science & Systems Specialty Group, Association of American Geographers* (2nd place; Graduate Student Awardee: Zhongfu Ma)	2025
Minnesota Student Digital Mapping Competition, *Minnesota Alliance for Geographic*	2025

Education (1st place; K-12 Student Awardee: De-An Chen)
Minnesota GIS/LIS Student Scholarship Award, *MN GIS/LIS Consortium* (Graduate Student Awardee: Meicheng Xiong) 2024
Robert Raskin Student Competition, *Cyberinfrastructure Specialty Group, Association of American Geographers* (2nd place; Graduate Student Awardee: Zhongfu Ma) 2023

RESEARCH, SCHOLARSHIP, AND CREATIVE WORK

Grants, Awards, Gifts, or Endowment Earnings

Awarded (**- internal source, **- external source, # - received at other institutions*)

*** The Structure of Human-Environment Interactions in Cities: Place, Network, and Evolution**
McKnight Land-Grant Professorship, University of Minnesota
PI: Zhu, Di; July 2026-June 2028; Funded Amount: $50,000

**** CTS Scholars, Center for Transportation Studies Operations**
Minnesota Department of Transportation
PI: Zhu, Di; September 2025 – June 2027; Funded Amount:$5,000

*** From Snowfall to Flow: GeoAI for Mobility Network Inference and Safer Winter Roads**
Grant-in-aid of Research, Artistry, and Scholarship program (GIA), University of Minnesota
PI: Zhu, Di; January 2026 - July 2027; Funded Amount: $39,575

*** Partnerships for Sustainable Communities (PSC) Course Grants**
Institute on the Environment, University of Minnesota
PI: Zhu, Di; September 2025 – July 2026; Funded Amount: $2,000

*** Rural Mobility and Access: Leveraging Big Data Analytics and Context-Aware Computing**
CTS Scholars Seed Research, Center for Transportation Studies, University of Minnesota
Co-PI: Zhu, Di, with PI: Ying Song; July 2024 - June 2025; Funded Amount: $74,478

*** Quantifying Carbon Emission Efficiency through the Lens of Human Mobility**
Medium/Large Seed Grant, Data Science Initiatives, University of Minnesota
PI: Zhu, Di; April 2024 - April 2025; Funded Amount: $10,000

*** Sensing Geospatial Communities in Mobility Networks: How Human Movements Drive Dynamic Community Structures within the Twin Cities Metro Area**
Faculty Interactive Research Program, Center for Urban & Regional Affairs (CURA)
PI: Zhu, Di; July 2022 – July 2023; Funded Amount: $43,757

*** A GeoAI-based Model for Human Mobility Flow Generation**
Seed Grants for Social Sciences Research, OFAA, College of Liberal Arts
PI: Zhu, Di; Mar 2023 – June 2024; Funded Amount: $4,250

 # Theoretical and analytical methods of spatial interaction networks in geospatial big data
The Major Program of the National Natural Science Foundation of China
Senior Personnel: Zhu, Di, with PI: Yu Liu, Co-PIs: Liyan Xu, Zhou Huang, Lun Wu**;** January 2019 - January 2021

 # Modeling spatial heterogeneity and spatial interactions from the big geo-data perspective
State Scholarship Fund of China, China Scholarship Council
PI: Zhu, Di; October 2018 - September 2019

Pending/Submitted **(**- internal source, **- external souce*)**

****A Cross-Scale Representation Approach to Sociodemographic Change-of-Support Problem**
Methodology, Measurement, and Statistics (MMS), National Science Foundation
PI: Zhu, Di; September 2026 - August 2029; Requested Amount: $475,561

****Identifying residential density thresholds to promote multimodal transportation in the Greater Twin Cities region**
Applied Research in Transportation (ART), Minnesota Department of Transportation
Co-PI: Zhu, Di, with PI: Jason Cao; July 2026 - June 2027; Requested Amount: $100,000

****World Models for Planetary Memory: Archives as Sensors of Climate Change**
Humanities and Artificial Intelligence Virtual Institute (HAVI), Schmidt Sciences
Co-PI: Zhu, Di, with PI: James Lee, Co-PIs: Rayvon Fouche, Han Liu, Kelsey Rydland, Yilun Du; September 2026 - August 2029; Requested Amount: $781,832

*** Mitigating Extreme Heat in Urban Systems: Integrating microclimate modeling, spatiotemporal data, and graph neural networks**
The University of Minnesota – Seoul National University Partnership Research Seed Grants, University of Minnesota
PI: Zhu, Di, with Co-PI: Steven Jige Quan; June 2026 - June 2027; Requested Amount: $40,000

Publications
(***** - *indicates corresponding author, Underline - indicates student/advisee author*)

Peer-Reviewed Journal Articles

Zhu, D.*, Wang, S., & Luo, P. (2026). Uncovering spatial process heterogeneity from graph-based deep spatial regression, *ISPRS Journal of Photogrammetry and Remote Sensing*, 232, 509-523. https://doi.org/10.1016/j.isprsjprs.2025.12.008
[Design (100%), Implementation (75%), Manuscript (80%)]

Cheng, X., Wang, T., **Zhu, D.**, & Ma, J. (2026). Uncertainty explanation of artificial intelligence models by SHAP. *Knowledge-Based Systems*, 115437. https://doi.org/10.1016/j.knosys.2026.115437
[Design (20%), Implementation (10%), Manuscript (20%)]

Du, J., Mugambi, T., **Zhu, D.**, Betway, B., & Odhiambo, N. (2026). Modelling uncertainty in data fusion: A knowledge graph approach. *Expert Systems with Applications*, 131550.

https://doi.org/10.1016/j.eswa.2026.131550
[Design (33%), Implementation (10%), Manuscript (10%)]

Wang, S., & **Zhu, D.*** (2025). A context-enhanced graph neural network operator for edge-to-edge learning in human mobility networks, *Transactions in GIS*, 29 (7): e70154. https://doi.org/10.1111/tgis.70154
[Design (80%), Implementation (10%), Manuscript (25%)]

Zhu, D.*, & Ma, Z. (2025). Gravity-informed deep flow inference for spatial evolution in panel data, *International Journal of Geographical Information Science*, 1-29. https://doi.org/10.1080/13658816.2025.2536512
[Design (80%), Implementation (40%), Manuscript (75%)]

Wang, S., & **Zhu, D.*** (2025). Inferring human movements after snowfall: Weather-informed graph learning model for flow redistribution in mobility networks, *International Journal of Geographical Information Science*, 1-29. https://doi.org/10.1080/13658816.2025.2524394
[Design (60%), Implementation (25%), Manuscript (25%)]

Xiong, M., & **Zhu, D.*** (2025). Lakeplace: Sensing interactions between lakes and human activities, *Environment and Planning B: Urban Analytics and City Science*, 23998083251386148. https://doi.org/10.1177/23998083251386148
[Design (60%), Implementation (20%), Manuscript (40%)]

Xiong, M., **Zhu, D.***, & David Van Riper (2025). A visitor-enriched census in the U.S. cities using large-scale mobile positioning data, *Scientific Data*, 12(1), 1106. https://doi.org/10.1038/s41597-025-05410-0
[Design (60%), Implementation (30%), Manuscript (33%)]

Ma, Z., & **Zhu, D.*** (2025). Collective flow evolution patterns reveal the mesoscopic structure in sequential spatial network snapshots. *International Journal of Geographical Information Science,* 39(1), 86-117. https://doi.org/10.1080/13658816.2024.2395953
[Design (90%), Implementation (40%), Manuscript (30%)]

Gao, S., Mai, G., Newsam, S., Yang, L., Lunga, D., **Zhu, D.**, Martins, B. and Arundel, S., (2025). Report of the 7th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery. *The SIGSPATIAL Special*, 15(1), 3-4. https://doi.org/10.1145/3757932.3757933
[Design (12.5%), Implementation (12.5%), Manuscript (12.5%)]

Zeng, X., Song, Y., & **Zhu, D.*** (2025). Spatially explicit knowledge in geo-embeddings: Interpreting location representation derived from human movement trajectories. *Transactions in Urban Data, Science, and Technology*, 27541231251333348. https://doi.org/10.1177/27541231251333348
[Design (30%), Implementation (20%), Manuscript (10%)]

Guo, H., Wang, H., **Zhu, D**., Fotheringham, S., & Liu, Y. (2025). RegionGCN: Spatial heterogeneity aware graph convolutional networks, *Annals of the American Association of Geographers*, 1-17. https://doi.org/10.1080/24694452.2025.2558661
[Design (33%), Implementation (25%), Manuscript (20%)]

Luo, P., Song, C., Li, H., **Zhu, D.**, & Duarte, F. (2025). Modeling shared micro-mobility as a

label propagation process for detecting overlapping communities. *Computers, Environment and Urban Systems*, 122, 102336. https://doi.org/10.1016/j.compenvurbsys.2025.102336
[Design (20%), Implementation (10%), Manuscript (10%)]

Wang, Y., Wang, Z., Zhang, F., Tang, C., Kang, C., **Zhu, D.**, Ma, Z., Ruan, S., Zhang, W., Zheng, Y., Yu, P., Liu, Y. (2025). A Gravity-informed spatiotemporal transformer for human activity intensity prediction, *Transactions on Pattern Analysis and Machine Learning*, 3625859. https://doi.org/10.1109/TPAMI.2025.3625859
[Design (30%), Implementation (20%), Manuscript (10%)]

Jiang, B., Cheng, T., Tsou, M. H., **Zhu, D.**, Ye, X. (2025). Advancing translational human dynamics research: bridging space, mind, and computational urban science in the era of GeoAI. *Computational Urban Science*, 5(1), 12. https://doi.org/10.1007/s43762-025-00171-3
[Design (20%), Implementation (20%), Manuscript (20%)]

Wang, S., Zhao, C., Jiang, Q., **Zhu, D.**, Ma, J., & Sun, Y. (2025). Application of graph convolutional neural networks and multi-sources data on urban functional zones identification, a case study of Changchun, China. *Sustainable Cities and Society*, 106116. https://doi.org/10.1016/j.scs.2024.106116
[Design (20%), Implementation (20%), Manuscript (15%)]

Wang, Y., & **Zhu, D.*** (2024). A hypergraph-based hybrid graph convolutional network for intracity human activity intensity prediction and geographic relationship interpretation. *Information Fusion*, 104, 102149. https://doi.org/10.1016/j.inffus.2023.102149
[Design (80%), Implementation (40%), Manuscript (60%)]

Zhang, Y., Yu, W., & **Zhu, D.** (2024). Next track point prediction using a flexible strategy of subgraph learning on road networks. *International Journal of Geographical Information Science*, 1-26. https://doi.org/10.1080/13658816.2024.2358527
[Design (30%), Implementation (10%), Manuscript (20%)]

Zhang, G., Gong, X., & **Zhu, D.** (2024). Geographic proximity and homophily effects drive social interactions within VGI communities: an example of iNaturalist. *International Journal of Digital Earth*, 17(1), 2297948. https://doi.org/10.1080/17538947.2023.2297948
[Design (40%), Implementation (20%), Manuscript (40%)]

Liu, Y., Wang, K., Xing, X., Guo, H., Zhang, W., Luo, Q., Gao, S., Huang, Z., Li, H., Li, X., Wang, J., Wang, J., & **Zhu, D**. (2023). On spatial effects in geographical analysis. *Acta Geographica Sinica*, 78(3), 517-531. https://doi.org/10.11821/dlxb202303001
[Design (20%), Implementation (0%), Manuscript (20%)]

Chen, T., **Zhu, D.**, Cheng, T., Gao, X., & Chen, H. (2023). Sensing dynamic human activity zones using geo-tagged big data in Greater London, UK during the COVID-19 pandemic. *PloS one*, 18(1), e0277913. https://doi.org/10.1371/journal.pone.0277913
[Design (30%), Implementation (40%), Manuscript (40%)]

Luo, P., Song, Y., **Zhu, D.**, Cheng, J., Meng, L. (2023). A generalized heterogeneity model for spatial interpolation. *International Journal of Geographical Information Science*, 37(3),

634-659. https://doi.org/10.1080/13658816.2022.2147530
[Design (20%), Implementation (20%), Manuscript (20%)]

Zhu, D.*, Liu, Y., Yao, X., & Fischer, M. M. (2022). Spatial regression graph convolutional neural networks: A deep learning paradigm for spatial multivariate distributions. *GeoInformatica*, 26 (4), 645-676. https://doi.org/10.1007/s10707-021-00454-x
[Design (80%), Implementation (70%), Manuscript (90%)]

Chen, T., Bowers, K., **Zhu, D.***, Gao, X., Cheng, T. (2022). Spatio-temporal stratified associations between urban human activities and crime patterns: a case study in San Francisco around the COVID-19 stay-at-home mandate. *Computational Urban Science*, 2(1), 13. https://doi.org/10.1007/s43762-022-00041-2
[Design (75%), Implementation (30%), Manuscript (30%)]

Zhang, Y., Yu, W., & **Zhu, D.** (2022). Terrain feature-aware deep learning network for digital elevation model superresolution. *ISPRS Journal of Photogrammetry and Remote Sensing*, 189, 143-162. https://doi.org/10.1016/j.isprsjprs.2022.04.028
[Design (20%), Implementation (10%), Manuscript (20%)]

Huang, X., **Zhu, D.**, Zhang, F., Liu, T., Li, X., & Zou, L. (2021). Sensing population distribution from satellite imagery via deep learning: Model selection, neighboring effects, and systematic biases. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 14, 5137-5151. https://doi.org/10.1109/JSTARS.2021.3076630
[Design (30%), Implementation (20%), Manuscript (20%)]

Zhu, D.*, Ye, X., & Manson, S. (2021). Revealing the spatial shifting pattern of COVID-19 pandemic in the United States. *Scientific Reports*, 11(1), 8396. https://doi.org/10.1038/s41598-021-87902-8
[Design (80%), Implementation (100%), Manuscript (80%)]

Xing, X., Huang, Z., Cheng, X., **Zhu, D.**, Kang, C., Zhang, F., & Liu, Y. (2020). Mapping human activity volumes through remote sensing imagery. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 13, 5652-5668. https://doi.org/10.1109/JSTARS.2020.3023730
[Design (10%), Implementation (10%), Manuscript (15%)]

Sari Aslam, N., **Zhu, D.**, Cheng, T., Ibrahim, M. R., & Zhang, Y. (2020). Semantic enrichment of secondary activities using smart card data and point of interests: A case study in London. *Annals of GIS*, 27(1), 29-41. https://doi.org/10.1080/19475683.2020.1783359
[Design (75%), Implementation (25%), Manuscript (35%)]

Zhu, D., Zhang, F., Wang, S., Wang, Y., Cheng, X., Huang, Z., & Liu, Y. (2020). Understanding place characteristics in geographic contexts through graph convolutional neural networks. *Annals of the American Association of Geographers*, 110(2), 408-420. https://doi.org/10.1080/24694452.2019.1694403
[Design (80%), Implementation (75%), Manuscript (80%)]

Yao, X., Gao, Y., **Zhu, D.**, Manley, E., Wang, J., & Liu, Y. (2020). Spatial origin-destination flow imputation using graph convolutional networks. *IEEE Transactions on Intelligent*

Transportation Systems, 22(12), 7474-7484. https://doi.org/10.1109/TITS.2020.3003310
[Design (25%), Implementation (25%), Manuscript (25%)]

Wang, Y., **Zhu, D.**, Yin, G., Huang, Z., & Liu, Y. (2020). A unified spatial multigraph analysis for public transport performance. *Scientific Reports*, 10(1), 9573. https://doi.org/10.1038/s41598-020-65175-x
[Design (50%), Implementation (25%), Manuscript (50%)]

Wu, L., Cheng, X., Kang, C., **Zhu, D.**, Huang, Z., & Liu, Y. (2020). A framework for mixed-use decomposition based on temporal activity signatures extracted from big geo-data. *International Journal of Digital Earth,* 13(6), 708-726. https://doi.org/10.1080/17538947.2018.1556353
[Design (10%), Implementation (10%), Manuscript (25%)]

Zhang, F., Zu, J., Hu, M., **Zhu, D.**, Kang, Y., Gao, S., Zhang, Y., & Huang, Z. (2020). Uncovering inconspicuous places using social media check-ins and street view images. *Computers, Environment and Urban Systems,* 81, 101478. https://doi.org/10.1016/j.compenvurbsys.2020.101478
[Design (10%), Implementation (25%), Manuscript (10%)]

Liu, Y., Yao, X., Gong, Y., Kang, C., Shi, X., Wang, F., Wang, J., Zhang, Y., Zhao, P., **Zhu, D.**, & Zhu, X. (2020). Analytical methods and applications of spatial interactions in the era of big data. *Acta Geographica Sinica*, 75(7), 1523-1538. https://doi.org/10.11821/dlxb202007014
[Design (10%), Implementation (10%), Manuscript (10%)]

Zhu, D., Cheng, X., Zhang, F., Yao, X., Gao, Y., & Liu, Y. (2020). Spatial interpolation using conditional generative adversarial neural networks. *International Journal of Geographical Information Science*, 34(4), 735-758. https://doi.org/10.1080/13658816.2019.1599122
[Design (100%), Implementation (75%), Manuscript (80%)]

Chen, L., Gao, Y., **Zhu, D.**, Yuan, Y., & Liu, Y. (2019). Quantifying the scale effect in geospatial big data using semi-variograms. *PloS one*, 14(11), e0225139. https://doi.org/10.1371/journal.pone.0225139
[Design (75%), Implementation (25%), Manuscript (40%)]

Zhang, F., Wu, L., **Zhu, D.**, & Liu, Y. (2019). Social sensing from street-level imagery: A case study in learning spatio-temporal urban mobility patterns. *ISPRS Journal of Photogrammetry and Remote Sensing*, 153, 48-58. https://doi.org/10.1016/j.isprsjprs.2019.04.017
[Design (50%), Implementation (33%), Manuscript (33%)]

Yao, X., Wu, L., **Zhu, D.**, Gao, Y., & Liu, Y. (2019). Visualizing spatial interaction characteristics with direction-based pattern maps. *Journal of Visualization*, 22, 555-569. https://doi.org/10.1007/s12650-018-00543-4
[Design (25%), Implementation (20%), Manuscript (10%)]

Zhang, S., **Zhu, D.***, Yao, X., Cheng, X., He, H., & Liu, Y. (2018). The scale effect on spatial interaction patterns: An empirical study using taxi OD data of Beijing and Shanghai. *IEEE Access*, 6, 51994-52003. https://doi.org/10.1109/ACCESS.2018.2869378

[Design (75%), Implementation (20%), Manuscript (50%)]

Yao, X., **Zhu, D.**, Gao, Y., Wu, L., Zhang, P., & Liu, Y. (2018). A stepwise spatio-temporal flow clustering method for discovering mobility trends. *IEEE Access*, 6, 44666-44675. https://doi.org/10.1109/ACCESS.2018.2864662

[Design (25%), Implementation (25%), Manuscript (25%)]

Zhu, D., Huang, Z., Shi, L., Wu, L., & Liu, Y. (2018). Inferring spatial interaction patterns from sequential snapshots of spatial distributions. *International Journal of Geographical Information Science*, 32(4), 783-805. https://doi.org/10.1080/13658816.2017.1413192

[Design (80%), Implementation (80%), Manuscript (80%)]

Liu, Y., Zhan, Z., **Zhu, D.**, Chai, Y., Ma, X., & Wu, L. (2018). Incorporating multi-source big geo-data to sense spatial heterogeneity patterns in urban space. *Geomatics and Information Science of Wuhan University*, 43(3), 327-335. https://doi.org/10.13203/j.whugis20170383

[Design (10%), Implementation (0%), Manuscript (10%)]

Zhu, D., Wang, N., Wu, L., & Liu, Y. (2017). Street as a big geo-data assembly and analysis unit in urban studies: A case study using Beijing taxi data. *Applied Geography*, 86, 152-164. https://doi.org/10.1016/j.apgeog.2017.07.001

[Design (50%), Implementation (100%), Manuscript (90%)]

Zhu, D.*, & Liu, Y. (2017). An incremental map-matching method based on road network topology. *Geomatics and Information Science of Wuhan University*, 42(1), 77-83. https://doi.org/10.13203/j.whugis20150016

[Design (50%), Implementation (100%), Manuscript (90%)]

Peer-Reviewed Book Chapters

Zhu, D.* & Ma, Z. (2025). Towards the Spatial Evolution between Distribution Snapshots: A Network Perspective Spatial Evolution. In *Urban Human Mobility - Practices, Analytics, and Strategies for Smart Cities,* edited by Huang, X., Ye, X., Stewart, K., and Das, S,. CRC Press, 177-192. https://doi.org/10.1201/9781003503262-18

[Design (80%), Implementation (50%), Manuscript (50%)]

Zhu, D.*, & Cao, G. (2024). Intelligent Spatial Prediction and Interpolation Methods. In *Handbook of Geospatial Artificial Intelligence (GeoAI),* edited by Gao, S., Hu, Y., and Li, W., CRC Press, 121-150. https://doi.org/10.1201/9781003308423-7

[Design (80%), Implementation (60%), Manuscript (80%)]

Zhu, D.*, & Hu, Y. (2023). Artificial Intelligence. In *Concise Encyclopedia of Human Geography,* edited by Lees, L., and Demeritt, D., Edward Elgar Publishing, 32-36. https://doi.org/10.4337/9781800883499.ch07

[Design (50%), Implementation (60%), Manuscript (50%)]

Peer-Reviewed Conference Proceedings

Wang, S., & **Zhu, D.*** (2025). A Deep Origin-Destination Flow Imputation Model Informed by the Visitation Law in Human Mobility. In Proceedings of *The 33rd ACM International Conference on Advances in Geographic Information Systems*, pp. 1266-1269. https://doi.org/10.1145/3748636.3766527

[Design (60%), Implementation (10%), Manuscript (20%)]

Song, Y., Zeng, X., Xiong, M., & **Zhu, D.** (2025). A Behavior-Informed and Geo-Context-Aware Home Detection Framework for Mobile Phone Positioning Data. In Proceedings of *The 33rd ACM International Conference on Advances in Geographic Information Systems*, pp. 1008-1018. https://doi.org/10.1145/3748636.3764171

[Design (30%), Implementation (20%), Manuscript (25%)]

Xiong, M., & **Zhu, D.*** (2025). A Graph-based Deep Population Downscaling Model on Irregular Spatial Units. In Proceedings of *The 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 37-44. https://dl.acm.org/doi/10.1145/3764912.3770815

[Design (50%), Implementation (20%), Manuscript (20%)]

Ma, Z., & **Zhu, D.*** (2025). A Neural Fitting Method for Potential Differential Equations in Urban Dynamics. In Proceedings of *The 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 45-48. https://doi.org/10.1145/3764912.3770816

[Design (60%), Implementation (20%), Manuscript (25%)]

Wang, S., & **Zhu, D.*** (2024). Edge Activating Module: Learning edge-to-edge features for mobility flow generation. In Proceedings of *The 7th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 11-14. https://doi.org/10.1145/3687123.369828

[Design (90%), Implementation (30%), Manuscript (20%)]

Xiong, M., & **Zhu, D.*** (2024). From Lake to Lakeplace: A Case Study of Lake-related Human Activities in the Twin Cities, In Proceedings of *The CaGIS+UCGIS Symposium*, pp. 1-7. https://cartogis.org/conferences/cagis2024/program

[Design (80%), Implementation (40%), Manuscript (30%)]

Zhu, D.*, Gao, S., Cao, G**.** (2022) Towards the Intelligent Era of Spatial Analysis and Modeling. In Proceedings of *The 5th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 10-13. https://doi.org/10.1145/3557918.3565863

[Design (90%), Implementation (80%), Manuscript (90%)]

Luo, P., & **Zhu, D.*** (2022). Sensing Overlapping Geospatial Communities from Human Movements using Graph Affiliation Generation Models. In Proceedings of *The 5th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 1-9. https://doi.org/10.1145/3557918.3565862

[Design (90%), Implementation (50%), Manuscript (40%)]

Wang, Y., & **Zhu, D.*** (2022) SHGCN: A Hypergraph-based Deep Learning Model for Spatiotemporal Traffic Flow Prediction. In Proceedings of *The 5th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 30-39. https://doi.org/10.1145/3557918.3565866

[Design (75%), Implementation (40%), Manuscript (40%)]

Zhang, W., Ma, Y., **Zhu, D.**, Dong, L., & Liu, Y. (2022). MetroGAN: Simulating Urban Morphology with Generative Adversarial Network. In Proceedings of *The 28th ACM*

SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 2482-2492. https://doi.org/10.1145/3534678.3539239
[Design (20%), Implementation (30%), Manuscript (20%)]

Chen, T., & **Zhu, D.*** (2021). The Spatio-temporal Stratified Association between Human Activities and Crime Patterns during the COVID-19 Stay-at-home Mandate. In Proceedings of *The 2021 ACM SIGSPATIAL China Annual Conference on Space Intelligence*, pp. 1-12. https://dx.doi.org/10.2139/ssrn.4004696
[Design (60%), Implementation (50%), Manuscript (60%)]

Chen, T., Cheng, T., & **Zhu, D.** (2021). The Exploration of Human Activity Zones using Geo-tagged Big Data during the COVID-19 First Lockdown in London, UK. In Proceedings of *The 29th Conference on GIS Research UK*, pp. 1-9. https://doi.org/10.5281/zenodo.4670050
[Design (60%), Implementation (30%), Manuscript (20%)]

Wang, Y., **Zhu, D.**, Yin, G., Huang, Z., & Liu, Y. (2019). Investigating Local Travel Speed with Spatial Network Structures and Properties. In Proceedings of *The 2nd International Conference on Urban Informatics*. https://www.polyu.edu.hk/lsgi/icui2019/programme/
[Design (50%), Implementation (25%), Manuscript (25%)]

Zhu, D.*, Cheng, T., & Liu, Y. (2019). Geo-propagation from Incomplete Spatial Distribution Data: A Case Study of House Price Estimation. In Proceedings of *The 27th Conference on GIS Research UK*, pp. 1-6. https://discovery.ucl.ac.uk/id/eprint/10085157.
[Design (80%), Implementation (100%), Manuscript (100%)]

Zhu, D.*, & Liu, Y. (2018). Modelling Spatial Patterns using Graph Convolutional Networks. In Proceedings of *The 10th International Conference on Geographic Information Science*, vol. 114, pp. 73(1)-73(7). https://doi.org/10.4230/LIPIcs.GISCIENCE.2018.73
[Design (90%), Implementation (100%), Manuscript (80%)]

Publications Online at ArXiv

Ma, Z., & **Zhu, D.*** (2026). Discovering governing spatial interaction mechanisms in dynamic urban systems. *arXiv preprint arXiv:2603.19537*. https://doi.org/10.48550/arXiv.2603.19537
[Design (75%), Implementation (10%), Manuscript (10%)]

Xiong, M., & **Zhu, D.*** (2025). Lakeplace: Sensing interactions between lakes and human activities. *arXiv preprint arXiv:2505.02289*. https://doi.org/10.48550/arXiv.2505.02289
[Design (60%), Implementation (20%), Manuscript (40%)]

Luo, P., & **Zhu, D**.* (2024). Uncover the nature of overlapping community in cities. *arXiv preprint: 2402.00222*. https://doi.org/10.48550/arXiv.2402.00222
[Design (90%), Implementation (40%), Manuscript (50%)]

Zhu, D.*, & Liu, Y. (2018). Modelling irregular spatial patterns using graph convolutional neural networks. *arXiv preprint arXiv:1808.09802*. https://doi.org/10.48550/arXiv.1808.09802
[Design (100%), Implementation (100%), Manuscript (90%)]

Publications Under Review

Luo, P., Xiong, M., Duarte, F., Santi, P., Ratti, C., & **Zhu, D.*** (2026). The spatial overlapping of communities explains urban mobility, *Nature Computational Science.*

Ma, Z., & **Zhu, D.*** (2026). Discovering governing spatial interaction mechanisms in dynamic urban systems. *Transactions in GIS.*

Chen, D., & **Zhu, D.*** (2026). Urban heat and green cover: Uncovering tree canopy's spatial effects on land surface temperature in the Minneapolis-St. Paul metropolitan area, *Computational Urban Science.*

Publications in Progress

Wang, S., & **Zhu, D.***, (2026). A survey of learning geographical relationships in GeoAI.

Wang, S., & **Zhu, D.***, (2026). Downscaling spatial origin-destination flows: A neural maximum entropy approach.

Harris, J., & **Zhu, D.***, (2026). EV-GCN-RL: Optimizing the Spatial Distribution of EV Chargers for Commuter Accessibility.

Xiong, M., & **Zhu, D.***, (2026). Deep areal interpolation of sociodemographic attributes.

Li, T., Pang, S., & **Zhu, D.**, (2026). Empirical coupling dynamics and graph-based spatiotemporal modeling of metro–bike integration: Evidence from Shanghai.

Zhu, D.*, (2026) GeoABC: A unified conceptual model for spatiotemporal issues in GIScience.

Zhu, D.*, (2026) The dawn of Geospatial Omics for comprehensive social sensing.

Code and Datasets

CEDGAN - Spatial interpolation using conditional generative adversarial neural networks. Released as free and open source repository on GitHub (21 forks and 63 stars): *https://github.com/dizhu-gis/cedgan-interpolation*

SRGCNN - A deep learning implementation of Spatial regression graph convolutional neural networks. Released as free and open source repository on GitHub (14 forks and 59 stars): *https://github.com/dizhu-gis/SRGCNN*

Visitor Census - A visitor-enriched census in the U.S. cities using large-scale mobile positioning data. Released as free and open dataset on GitHub: *https://github.com/GeoDI-Lab/Visitor_census*

HyGCN - Hypergraph-based Hybrid Graph Convolutional Network. Released as free and open source repository on GitHub (1 fork and 23 stars): *https://github.com/GeoDI-Lab/HyGCN*

EAGO - A Plug-and-Play Edge-Activating Graph Operator. Released as free and open source repository on GitHub: *https://github.com/GeoDI-Lab/EAGO-TGIS*

DSEN - Deep Spatial Evolution Network. Released as free and open source repository on GitHub: *https://github.com/GeoDI-Lab/DSEN-IJGIS*

WIMN - Weather-Informed Mobility Network. Released as free and open source repository on GitHub (1 fork and 4 stars): *https://github.com/GeoDI-Lab/WIMN-IJGIS*

CFEP - Detect collective flow evolution patterns from spatial network snapshots. Released as free and open source repository on GitHub: *https://github.com/GeoDI-Lab/CFEP-Collective_Flow-Evolutionary_Pattern*

Presentations, Posters, and Exhibits
*(Underline - indicates student co-presenter, * - indicates advisor)*

Keynote/Panelist

Zhu, D., "GeoAI and Spatial Networks for Urban Social Sensing", *Panelist*, Bureau of Transportation Statistics (BTS): National Transportation Atlas Database and Geospatial Transportation Programs Expert Meeting, National Academies of Science, Engineering, and Medicine (NASEM), Washington D.C., US. (Apr 20, 2026).

Zhu, D. "When Human Dynamics meets GeoAI", *Panelist*, the 10th Symposium on Human Dynamics Research, Annual Meeting of American Association of Geographers (AAG), Hawaii, US. (Apr. 18, 2024).

Zhu, D. "Sensing the overlapping nature of mobility networks using geospatial artificial intelligence", *Panelist*, Symposium on Replicable Spatiotemporal Data Science, Spatial Data Lab, Center for Geographic Analysis, Harvard University, US. (Jul 15-16, 2023).

Zhu, D. "Artificial Intelligence in Urban Science", *Keynote*, Opening webinar on Urban Science, Swarma Club. (Jul 1, 2023).

Zhu, D. "Intelligent spatial prediction: Rethinking geospatial modeling in the era of GeoAI", *Keynote*, Annual Conference of Geomatics and GIScience, Central South University, China. (Dec 26, 2020).

Conference Presentations

Zhu, D.* (2026). *EAGO: A Context-Enhanced GNN Operator for Edge-to-Edge Learning in Mobility Networks*. Mar. 16-21, San Francisco, CA, United States (AAG 2026)

Harris, J., & **Zhu, D.*** (2026). *EV-GCN-RL: Spatial Optimization of EV Charger Accessibility*. Mar. 16-21, San Francisco, CA, United States (AAG 2026)

Ma, Z., & **Zhu, D.*** (2026). *Discovering Governing Spatial Interaction Mechanisms in Dynamic Urban Systems*. Mar. 16-21, San Francisco, CA, United States (AAG 2026)

Xiong, M., & **Zhu, D.*** (2026). *A graph-based deep population downscaling model on irregular spatial units*. Mar. 16-21, San Francisco, CA, United States (AAG 2026)

Chen, D., & **Zhu, D.*** (2026). *Urban Heat and Green Cover: Uncovering Tree Canopy's Spatial Effects on Land Surface Temperature in the Minneapolis-St. Paul Metropolitan Area*. Mar. 16-21, San Francisco, CA, United States (AAG 2026)

Zhu, D.* (2025). *A visitor-enriched census in the U.S. cities using large-scale mobile positioning data*. Oct. 22-25, Minneapolis, MN, United States (ACSP 2025)

Zeng, X., Song, Y., Xiong, M., & **Zhu, D** (2025). *Rural Mobility and Access: Leveraging Big Data Analytics and Context-Aware Computing*. Nov. 6, Minneapolis, MN, United States (CTS Transportation Research Conference 2025)

Song, Y., Xiong, M., Zeng, X., & **Zhu, D**. *A Behavior-Informed and Geo-Context-Aware Home Detection Framework for Mobile Phone Positioning Data*. Nov. 3-6, Minneapolis, MN, United States (SIGSPATIAL 2025)

Zhu, D.* (2025). *A visitor-enriched census in the U.S. cities using large-scale mobile positioning data*. Oct. 1-3, Duluth, MN, United States (MN GIS/LIS Annual Conference 2025)

Zhu, D.* (2025). *RegionGCN: Spatial-Heterogeneity-Aware Graph Convolutional Networks*, Mar. 24-28, Detroit, MI, United States (AAG 2025).

Ma, Z., & **Zhu, D.*** (2025). *Gravity-Informed Deep Flow Inference for Spatial Evolution in Panel Data*, Mar. 24-28, Detroit, MI, United States (AAG 2025).

Harris, J., & **Zhu, D.*** (2025). *Multi-City Urban Scaling: Insights from Human Mobility Data on Carbon Emissions Efficiency*, Mar. 24-28, Detroit, MI, United States (AAG 2025).

Wang, S., & **Zhu, D.*** (2024). *Edge Activating Module: Learning edge-to-edge features for mobility flow generation*. Oct. 29-Nov. 1, Atlanta, GA, United States (GeoAI'24).

Zhu, D.* (2024). *A hypergraph-based hybrid graph convolutional network for human activity intensity prediction*, Apr. 16-20, Hawaii, United States (AAG 2024)

Zhu, D.* (2024). *From Lake to Lakeplace: A Case Study of Lake-related Human Activities in the Twin Cities***,** Jun. 3-6, Ohio, Columbus, United States (CaGIS+UCGIS Symposium 2024).

Xiong, M., & **Zhu, D.*** (2024). *Lakeplace: Sensing human-lake interactions with mobile phone data*, Apr. 16-20, Hawaii, United States (AAG 2024)

Ma, Z., & **Zhu, D.*** (2024). *Bi-directional spatial evolution networks for flow inference*, Apr. 16-20, Hawaii, United States (AAG 2024)

Wang, S., & **Zhu, D.*** (2024). *Generating human mobility network under extreme snowfall: A case study in the winter of Twin Cities*, Apr. 16-20, Hawaii, United States (AAG 2024)

Ma, Z., **Zhu, D.*** (2023). *Collective Flow Evolution as a Mesoscopic Structure of Spatial Network, July*. 20-22, London, United Kingdom (GeoInformatics 2023)

Zhu, D.* (2023). *Learning Spatial Heterogeneity via Explainable Deep Spatial Regression*. Mar. 23-27, Denver, United States (AAG 2023)

Ma, Z., **Zhu, D.*** (2023). *Collective Flow Evolution Pattern: A mesoscopic exploration of spatial network dynamics*, Mar. 23-27, Denver, United States (AAG 2023)

Hendrickson, R., **Zhu, D.*** (2023). *Exploring the Scaling Relationships between Human Mobility and Air Pollutants in the Twin Cities*, Mar. 23-27, Denver, United States (AAG 2023)

Zhu, D.*, Liu, Y., Yao, X., & Fischer, M.M. (2022). *Spatial Regression Graph Neural Networks*, Feb. 26, Virtual (AAG 2022)

Zhu, D.*, Zhang, F., Cheng, X., & Liu, Y. (2019). *Spatial interpolation based on conditional generative adversarial neural networks*, Apr. 3-7, Washington, DC, United States (AAG 2019)

Xing, X., **Zhu, D.**, Cheng, X., & Liu, Y. (2018). *Population mapping based on deep features*

of remote sensing imagery, Jun. 28-30, Kunming, China. (GeoInformatics 2018)

Chen, L., **Zhu, D.**, & Liu, Y. (2018). *Quantify the scale effect in geospatial big data using semi-variograms*, Jun. 28-30, Kunming, China. (GeoInformatics 2018)

Zhu, D., Shi, L., Wang, Y., Cheng, X., & Liu, Y. (2017). *Infer spatial interaction patterns from spatial distributions*, Aug. 2-4, Buffalo, United States (GeoInformatics 2017)

Zhu, D. (2016). *A map visualization of the air quality index in China*, Jul. 23, Changsha, China (ChinaVis 2016)

Zhu, D., Wang, N., & Liu, Y. (2016). *Street perspective: a novel spatial unit in urban social sensing*, Aug. 18-20, Beijing, China (SDH 2016)

Zhu, D., & Liu, Y. (2016). *The distance effect in spatial interaction and spatial similarity: a big data view of Tobler's First Law*, Aug. 21-23, Beijing, China (IGC 2016)

Invited Presentations at Colloquia and Seminars

Zhu, D., "Spatial Networks and AI for Social Sensing", Geospatial Data Science Seminar, Department of Geography, University of Wisconsin-Madison. (Apr 10, 2026). *Invited*

Zhu, D., "Spatial Networks & GeoAI for Urban Social Sensing", GIS Speaker Series, University of Minnesota, Twin Cities. (Feb 19, 2026). *Invited*

Zhu, D., "Spatial Network and Panel Perspective in GeoAI", Workshop on Critical Challenges of GeoAI, Spatial Analysis Research Center (SPARC), Arizona State University. (Feb 8, 2026). *Invited*

Zhu, D., "Geospatial Data Intelligence for Spatial Prediction", Urban Informatics Series, Department of Urban Informatics, Shenzhen University. (May 16, 2024). *Invited*

Zhu, D., "Geospatial Data Intelligence for Human-Environment Systems ", Data Management Lab, Department of Computer Science and Engineering, University of Minnesota. (May 22, 2024). *Invited.*

Zhu, D., "Intelligent Spatial Prediction and Interpolation Methods ", GeoAI Solutions for Sustainable Development: The Handbook of Geospatial Artificial Intelligence (GeoAI)', AI for Good webinar, International Telecommunication Union (ITU), United Nations (Feb 23, 2024). *Invited*.

Zhu, D. "Geospatial Data Intelligence for Social Sensing", Coffee Hour Lecture Series, Department of Geography, Environment, and Society, University of Minnesota, Twin Cities. (Nov 9, 2023). *Invited.*

Zhu, D. "Geospatial Artificial Intelligence", Series webinar on Urban Science, Swarma Club. (Aug 5, 2023). *Invited*

Zhu, D. "The overlapping community structure and collective flow evolution pattern in urban mobility networks", Department of Intelligent Engineering, Sun Yat-Sen University. (Jul 4, 2023). *Invited*

Zhu, D. "The overlapping community structure and collective flow evolution pattern in urban mobility networks", Department of Geography, Northeast Normal University. (Jun 30, 2023). *Invited*

Zhu, D. "The overlapping community structure and collective flow evolution pattern in

urban mobility networks", Department of Geography, Beijing Normal University. (Jun 28, 2023). *Invited*

Zhu, D. "The overlapping community structure and collective flow evolution pattern in urban mobility networks", School of Earth and Space Science, Peking University. (Jun 27, 2023). *Invited*

Zhu, D. "The overlapping community structure and collective flow evolution pattern in urban mobility networks", Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Science. (Jun 26, 2023). *Invited*

Zhu, D. "The overlapping community structure and collective flow evolution pattern in urban mobility networks", Department of Geoscience and Information Physics, Central South University. (Jun 25, 2023). *Invited*

Zhu, D. "How human movements drive complex and dynamic community structures within the Twin Cities Metro", Minnesota Compass, Amherst H. Wilder Foundation. (Apr 5, 2023). *Invited*.

Zhu, D. "Revealing the flow patterns underlying spatial distribution snapshots", Humanistic GIS Lab, Department of Geography, University of Washington, Seattle. (Oct 31, 2022). *Invited*.

Zhu, D. "Network-based GeoAI analytics," Research Seminar at the Chair of Cartography and Visual Analytics, School of Engineering and Design/Department of Aerospace and Geodesy, Technical University Munich. (Jul 21, 2022). *Invited*.

Zhu, D. "Intelligent spatial prediction in incomplete-data scenarios," GeoAI Research Seminar Discussion Knowledge Computing Lab, Department of Computer Science & Engineering, University of Minnesota. (Dec 16, 2021). *Invited*.

Zhu, D. "Inferring spatial interaction pattern from spatial distribution snapshots," China Data Lab 2021 Workflow Webinar, Wuhan University & Harvard University. (Jun 18, 2021). *Invited*.

Zhu, D. "Intelligent spatial understanding: representation, modeling and prediction," CPGIS 2021 GeoAI Seminar Series, China University of Geosciences. (May 9, 2021). *Invited*.

Zhu, D. "Linkages between Spatial Regression and Graph Neural Networks," The 5th GIScience Symposium, Peking University, Beijing Normal University, and Chinese Academy of Science. (Apr 17, 2021). *Invited*.

Zhu, D. "Inferring national migration flows from sequential population snapshots," Geospatial Seminar, Department of Civil Environmental & Geomatic Engineering, University College London. (Feb 21, 2019). *Invited*

Guest Lecture

Zhu, D., “Spatial Networks and GeoAI for Urban Social Sensing”, ARCH 5721: Future Cities as a Frontier of Urban Design, College of Design, University of Minnesota, Twin Cities. (Apr 28, 2026).

Zhu, D. “Human & Environment Interaction in Cities: Place, Network, and Evolution”,

GEOG 8001: Problems in Geographic Thought, University of Minnesota, Twin Cities. (Sep 22, 2025).

Zhu, D. "When GeoAI meets Human Environment", GEOG 8001: Problems in Geographic Thought, University of Minnesota, Twin Cities. (Nov 14, 2024).

Zhu, D. "Artificial Intelligence + Geospatial", GEOG 8001: Problems in Geographic Thought, University of Minnesota, Twin Cities. (Nov 30, 2023).

Zhu, D. "Intelligent Sensing of Urban Space in Street Perspective," Intelligent Transportation Systems, Institute of Remote Sensing and GIS, Peking University. (Oct 17, 2017).

Posters or Exhibitions

Wang, S., & **Zhu, D.*** (2025). A Deep Origin-Destination Flow Imputation Model Informed by the Visitation Law in Human Mobility. Poster Exhibition In *The 33rd ACM International Conference on Advances in Geographic Information Systems* (SIGSPATIAL '25), Nov., Minneapolis, MN, US. (Nov. 5, 2025)

Xiong, M., **Zhu, D.***, Van Riper, D., "A visitor-enriched census in the U.S. cities using large-scale mobile positioning data", Research Computing Exhibition, Minnesota Supercomputing Institute, University of Minnesota, Twin Cities, Minneapolis, MN, US (Apr. 24, 2025)

Zhu, D.* "About spatial heterogeneity patterns," Academic Star Competition, School of Earth and Space Sciences, Peking University, Feb., Beijing, China (Feb 26, 2017)

TEACHING

University of Minnesota

GEOG 3531/5531 Principles of Spatial Analysis — 2025 - Present
Fall 2025, Spring 2026

GEOG 3531/5531 Numerical Spatial Analysis — 2021 - 2025
Fall 2021, Spring 2022, Fall 2022, Fall 2023, Spring 2025

GEOG 3561/5561 Principles of GIS — 2026
Spring 2026

GEOG 5533 Advanced Spatial Analysis — 2024 -Present
Spring 2024

GIS 5555 Basic Spatial Analysis — 2023 - Present
Fall 2023, Spring 2025

GEOG 8295 Seminar in Geospatial Artificial Intelligence — 2022 - Present
Spring 2022, Spring 2023, Spring 2024

GEOG 1502 Mapping Our World — 2025
Fall 2025 (co-taught)

Peking University

04831410: Introduction to Computation — 2016 - 2018
Teaching Assistant

01213660: Intelligent Transportation Systems 2017
Guest Lecturer
01213610: GIS Algorithms and Applications 2016 - 2017
Teaching Assistant
01235080: Geo-mathematical Models 2015
Teaching Assistant

University College London
CEGE 0097 Spatial Analysis and Geocomputation 2019
Guest Lecturer, Tutor

CURRICULUM DEVELOPMENT

Curriculum Development Activities

Redesigning Spatial Analysis curriculum across multiple courses	2021 - 2025
New Faculty Program, University of Minnesota	2022
Early Career Teaching and Learning Program, University of Minnesota	2024, 2025

ADVISING AND MENTORING

Graduate Student Activities

Advisees

Jacob Harris, *Geography MA, PhD*	Advisor	2024 - Present
Meicheng Xiong, *Geography PhD*	Advisor	2023 - Present
Sheng Wang, *Geography PhD*	Advisor	2023 - Present
Zhongfu Ma, *Geography PhD*	Advisor	2022 - Present
Ziliang Jin, *Master of GIS*	Advisor	2025 - 2026
Rob Hendrickson, *Master of GIS*	Advisor	2022 - 2023
Gene (Ziying) Cheng, *Master of GIS*	Advisor	2022 - 2023

Doctoral Preliminary Exam Committee Member

Shuai An, *CSE PhD Preliminary*	Committee member	2025
Xiaohuan Zeng, *Geography PhD Preliminary*	Committee member	2025
Youssef Hussein, *CSE PhD Preliminary*	Committee member	2025
Mohamed Hemdan, *CSE PhD Preliminary*	Committee member	2025
Mohsen Ahmadkhani, *Geography PhD Preliminary*	Committee member	2022
Zekun Li, *CSE PhD Preliminary*	Committee member	2022

Doctoral Final Committee

Tianyi Li, *CEGE PhD Defense*	Committee member	2024
Mohsen Ahmadkhani, *Geography PhD Defense*	Committee member	2024
Zekun Li, *CSE PhD Defense*	Committee member	2026
Xiaohuan Zeng, *Geography PhD Defense*	Committee member	2026

Masters Thesis/Research Committee

Dale Thuftedal, *Geography MA Defense*	Committee chair	2026
Xiaohuan Zeng, *Geography MA Defense*	Committee member	2022 - 2023
Nithya Murikinati, *Data Science, MS Defense*	Committee member	2025

Other Advising Activities

Ankila Kumari, *Master of GIS*	Career Advising	2025
Zain Ul Abdin Siyal, *Master of GIS*	Career Advising	2024
Peng Luo, *PhD, Massachusetts Institute of Technology*	Peer Mentoring	2023-2026
Tongxin Chen, *PhD*, *University College London*	Peer Mentoring	2020-2024
Yi Wang, *PhD, Peking University*	Peer Mentoring	2023-2025
Tianyi Li, *PhD, University of Minnesota*	Peer Mentoring	2023-2025

Undergraduate Student Activities

DFRACS (Dean's First-Year Research & Creative Scholars)

Shawn Li, *Mathematics*	DFRACS Mentor	2025 - 2026
Ryan Mitchell, *Biology, Society, and Environment*	DFRACS Mentor	2025 - 2026
Mathias Kreuziger, *Biology, Society, and Environment*	DFRACS Mentor	2024 - 2025
Sumeiya Mohamed, *Computer Science*	DFRACS Mentor	2024 - 2025

UROP (Undergraduate Research Opportunities Program)

Mathias Kreuziger, *Biology, Society, and Environment*	UROP Mentor	2026

(Mathias' project: "Examining Isolation and Human Mobility on Minnesota's Native American Reservations" received the UROP award in 2025)

K-12 Student Activities

Advising Activities

De-An Chen, *St. Paul Academy and Summit School*	Research Mentoring	2025 - 2026

De-An's project:"2025 Plan to Reduce Urban Heat in Minneapolis," won First Place out of 126 entries in the Minnesota Student Digital Mapping Competition, hosted by the Minnesota Alliance for Geographic Education (MAGE)

SERVICE AND PUBLIC OUTREACH

Service to the Discipline/Profession/Interdisciplinary Area(s)

Editorship/Guest Editorship

Associate Editor	*Computational Urban Science*	2025 - Present
Editorial Board	*Humanities and Social Sciences Communications*	2025 - Present
Guest Editor	*Annals of GIS*	2026

Guest Editor	*Computational Urban Science*	2024 - 2025
Guest Editor	*Frontiers in Environmental Science*	2022
Guest Editor	*Remote Sensing*	2022

Reviewer

Spatial Statistics	2016 - Present
Acta Geodaetica et Cartographica Sinica	2017 - Present
Computers, Environment, and Urban Systems	2017 - Present
Applied Geography	2018 - Present
IEEE ACCESS	2018 - Present
IEEE Transactions on Industrial Informatics	2018 - Present
International Journal of Geographical Information Science	2018 - Present
PLOS ONE	2018 - Present
Annals of the American Association of Geographers	2020 - Present
Cities	2020 - Present
Computational Urban Science	2020 - Present
Geo-spatial Information Science	2020 - Present
Nature Scientific Reports	2020 - Present
Transactions on Spatial Algorithms and Systems	2020 - Present
Cartography and Geographic Information Science	2021 - Present
CRC Press - Taylor & Francis	2021 - Present
Geomatics and Information Science at Wuhan University	2021 - Present
Humanities and Social Sciences Communications	2021 - Present
International Journal of Applied Earth Observation and Geoinformation	2021 - Present
International Journal of Digital Earth	2021 - Present
ISPRS International Journal of Geo-Information	2021 - Present
Journal of Planning Education and Research	2021 - Present
Stochastic Environmental Research and Risk Assessment	2021 - Present
Artificial Intelligence in Geosciences	2022 - Present
IEEE Geoscience and Remote Sensing Letters	2022 - Present
ISPRS Journal of Photogrammetry and Remote Sensing	2022 - Present
Journal of Transport and Land Use	2022 - Present
Transactions in GIS	2022 - Present
Environment and Planning B: Urban Analytics and City Science	2023 - Present
Geographical Analysis	2023 - Present
Scientific Data	2024 - Present
Sustainable Cities and Society	2024 - Present
Travel Behaviour and Society	2025 - Present
ISPRS Congress	2025 - Present
Journal of Geographical Systems	2026 - Present

Review panels for external funding agencies, foundations, etc.

National Science Foundation, Methodology, Measurement, and Statistics (MMS)	Reviewer	2025
National Science Foundation, Computer and Information Science and Engineering (CISE), Information & Intelligent Systems (IIS)	Panel Reviewer	2023
National Science Foundation, Human-Environment and Geographical Sciences (HEGS)	Reviewer	2023
National Science Foundation, Environmental Sustainability (ES)	Reviewer	2023

Organization of Conferences, Workshops, Panels, Symposia

The 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL' 26)	Local chair (outgoing)	2026
AAG 2026 Symposium on Spatial AI and Data Science	PC Member	2026
AAG 2026 Symposium on Spatial AI and Data Science: AI and Spatial Networks	Session Chair	2026
AAG 2026 Symposium on GeoAI and Deep Learning: GeoAI for Spatial Analytics and Modeling	Session Chair	2026
AAG 2026: Geographic Principles on Spatial Analysis and Modeling	Session Co-Organizer	2026
Institute for Geospatial Understanding through an Integrative Discovery Environment Forum 2026 (I-GUIDE'26)	PC Member	2026
The 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026)	PC Member	2026
The 33rd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL' 25)	Local chair	2025
The 8th International Workshop on AI for Geographic Knowledge Discovery (GeoAI'25)	PC Chair	2025
AAG 2025 Symposium on GeoAI and Deep Learning: GeoAI for Spatial Analytics and Modeling,	Session Chair	2025
Geo for AI vs. AI for Geo, International Symposium on Spatiotemporal Data Science: GeoAI for Social Sciences (ISSDS 2024)	Panel Chair	2024
Institute for Geospatial Understanding through an Integrative Discovery Environment Forum 2024 (I-GUIDE'24)	PC Member	2024

The 7th International Workshop on AI for Geographic Knowledge Discovery (GeoAI'24)	PC Chair	2024
AAG 2024 Symposium on GeoAI and Deep Learning: GeoAI for Spatial Analytics and Modeling	Session Chair	2024
The 2024 International Symposium on Spatiotemporal Data Science (ISSDS 2024)	PC Member	2024
The 30th International Conference on GeoInformatics (GeoInformatics'23)	PC Member	2023
The 2nd International Workshop on Geospatial Knowledge Graphs and GeoAI: Methods, Models, and Resources (GeoKG & GeoAI 2023)	PC Member	2023
Institute for Geospatial Understanding through an Integrative Discovery Environment Forum 2023 (I-GUIDE'23)	PC Member	2023
AAG 2022 Symposium on Data-Intensive Geospatial Understanding: Intelligent Geospatial Analytics	Session Chair	2022
The 28th Geographical Information Science Research UK Conference (GISRUK'20)	Session Chair	2020
The 5th International Workshop on AI for Geographic Knowledge Discovery (GeoAI'22)	PC Member	2022
The 4th International Workshop on AI for Geographic Knowledge Discovery (GeoAI'21)	PC Member	2021
The 3rd International Workshop on AI for Geographic Knowledge Discovery (GeoAI'19)	PC Member	2019

Service to the University/College/Department

Department

Member, Merit Committee	2025 - 2026
Member, Admissions Committee	2022 - 2024
Member, Awards Committee	2022 - 2024
Member, Graduate Education Policy Committee	2022 - 2023
Member, Coffee Hour Committee	2022 - 2023
Member, Undergraduate Education Policy Committee	2021, 2025, 2026

College

Member, CLA Assembly	2026 - 2028

University

Member, Executive Committee, Master of GIS (MGIS) Program	2022 - Present
Member, Center for Transportation Studies	2024 - Present

Public and External Service

Member, Research Committee, University Consortium for Geographic Information Science (UCGIS)	2026 - Present
Member, Conference Site Selection Committee, International Association of Chinese Professionals in Geographic Information Sciences (CPGIS)	2024 - Present
Member, Career Development Committee, International Association of Chinese Professionals in Geographic Information Sciences (CPGIS)	2022 - Present
Judge, John Odland Student Competition of Spatial Analysis & Modeling (SAM) Specialty Group at Association of American Geographers (AAG)	2025, 2026
Judge, MinneMudac Student Data Science Competition, MinneAnalytics	2025
Member, Election Committee, International Association of Chinese Professionals in Geographic Information Sciences (CPGIS)	2024, 2025
BOD Member, International Association of Chinese Professionals in Geographic Information Sciences (CPGIS)	2023 - 2025